A NONZERO IDEAL IN AN ALGEBRAIC EXTENSION OF DOMAINS MEETS THE SUBRING NONTRIVIALLY

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Algebraicity over the fraction field gives a polynomial relation. Used inside the counted proof at lines 6187-6190. Produces a relation with coefficients in A and nonzero constant term for a chosen element of the extension.
- The constant term of a relation lies in the ideal. Used at the last line of the counted proof (line 6191). Combines with divisibility of the leading term by x to exhibit a nonzero element of A inside J.

Lemma 0.1. *Let $A \subset B$ be an inclusion of domains inducing an algebraic extension of fraction fields. If $J \subset B$ is a nonzero ideal, then $A \cap J$ is nonzero too. Thus the image of a proper closed subset of* $\mathrm{Spec}(B)$ *is not dense in* $\mathrm{Spec}(A)$.

Proof. Let $x \in J$ be a nonzero element. Since x is algebraic over the fraction field of A, there exists a $d \geq 1$ and $a_0, \ldots, a_d \in A$ with $a_0, a_d \neq 0$ such that $a_d x^d + a_{d-1}x^{d-1} + \ldots + a_0 = 0$ in B. Then $a_0 \in A \cap J$. □

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